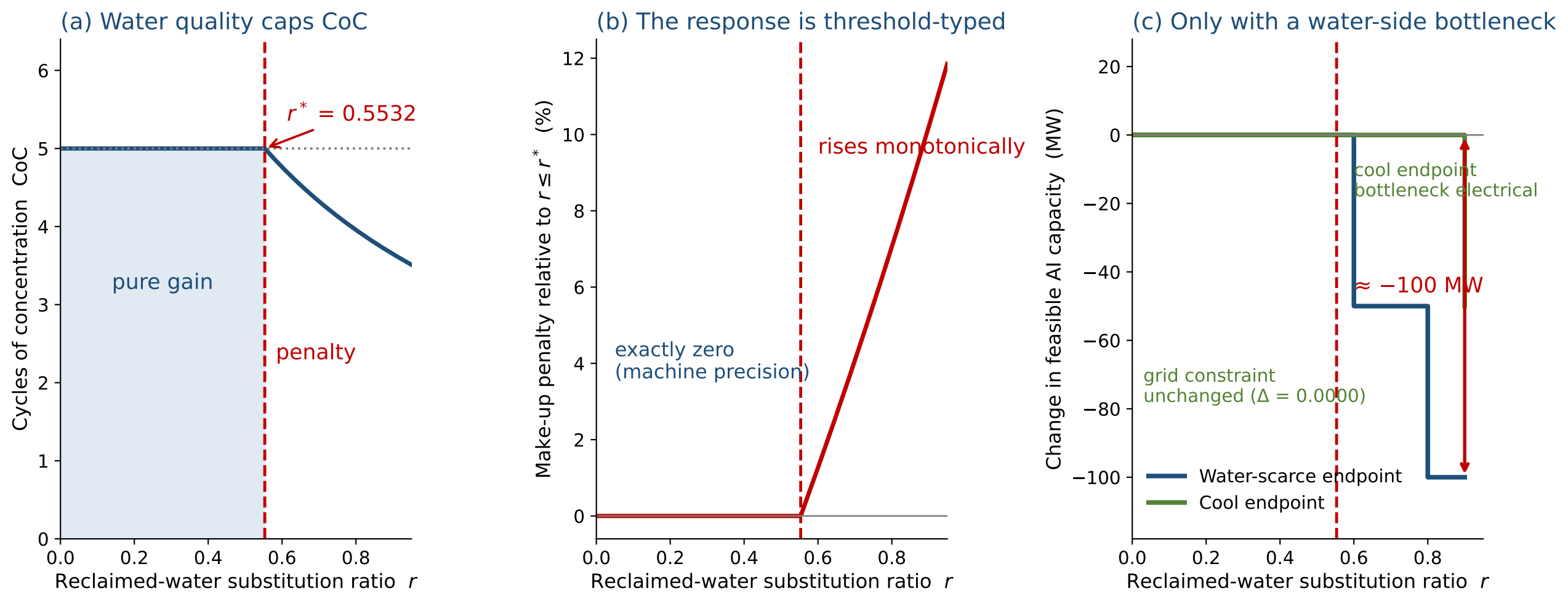
(a) Water quality caps CoC
Cycles of concentration CoC
Reclaimed-water substitution ratio r
r* = 0.5532
pure gain
penalty
(b) The response is threshold-typed
Make-up penalty relative to r ≤ r* (%)
Reclaimed-water substitution ratio r
rises monotonically
exactly zero
(machine precision)
(c) Only with a water-side bottleneck
Change in feasible AI capacity (MW)
Reclaimed-water substitution ratio r
cool endpoint
bottleneck electrical
≈ −100 MW
grid constraint
unchanged (Δ = 0.0000)
Water-scarce endpoint
Cool endpoint